Introduction to Data Mining

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The slides have been modified and rearranged by G.A. Putri Saptawati for the purpose of course IF4041 Data Mining

Main Reference

Jianwei Han, Micheline Kamber, Jian Pei: “Data Mining: Concepts and Techniques”, 4th Edition, Chapter 1. Introduction, Elsevier Inc., 2023

Chapter 1. Introduction

- What Is Data Mining?
- Data Mining: An Essential Step in Knowledge Discovery
- Diversity of Data Types for Data Mining
- Mining Various Kinds of Knowledge
- Data Mining: Confluence of Multiple Disciplines
- Data Mining and Applications
- Data Mining and Society
- Summary

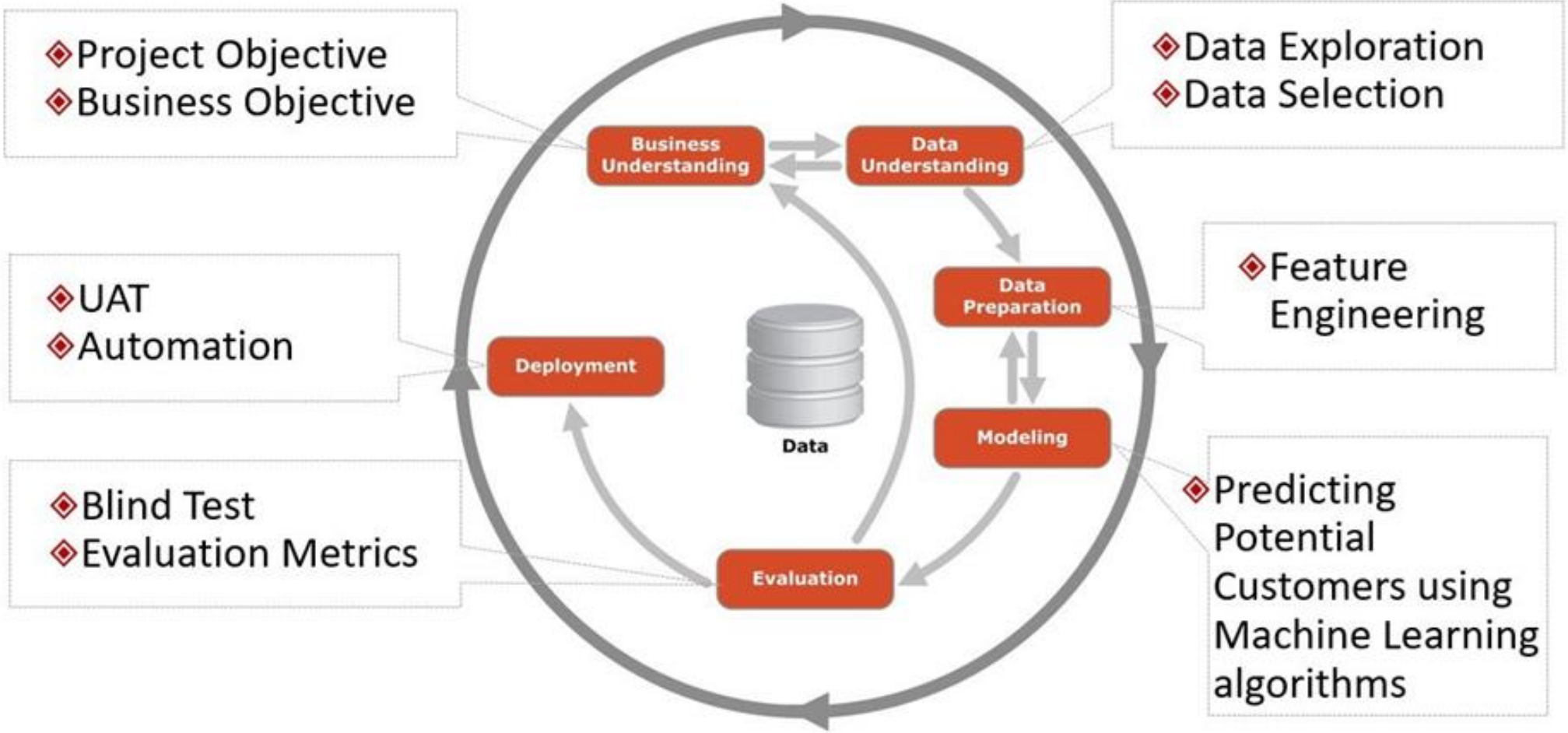
What Is Data Mining?

- We live in a world where vast amounts of data are generated constantly and rapidly
- **Data mining** is the process of discovering interesting patterns, models and other kinds of knowledge in large data sets
  - "Data mining": a misnomer? It should be "knowledge mining from data"
  - Other terms: *Knowledge mining from data*, *KDD* (*Knowledge Discovery from Data*), *pattern discovery*, *knowledge extraction*, *data analytics*, *information harvesting*
- Data mining is a young, dynamic, and promising field
- Example: Data mining turns a large collection of data into knowledge
  - Google's *Flu Trends* found a close relationship between the number of people who search for flu-related info. and the number of people who have flu symptoms
    - It can estimate flu activity up to two weeks faster than traditional systems

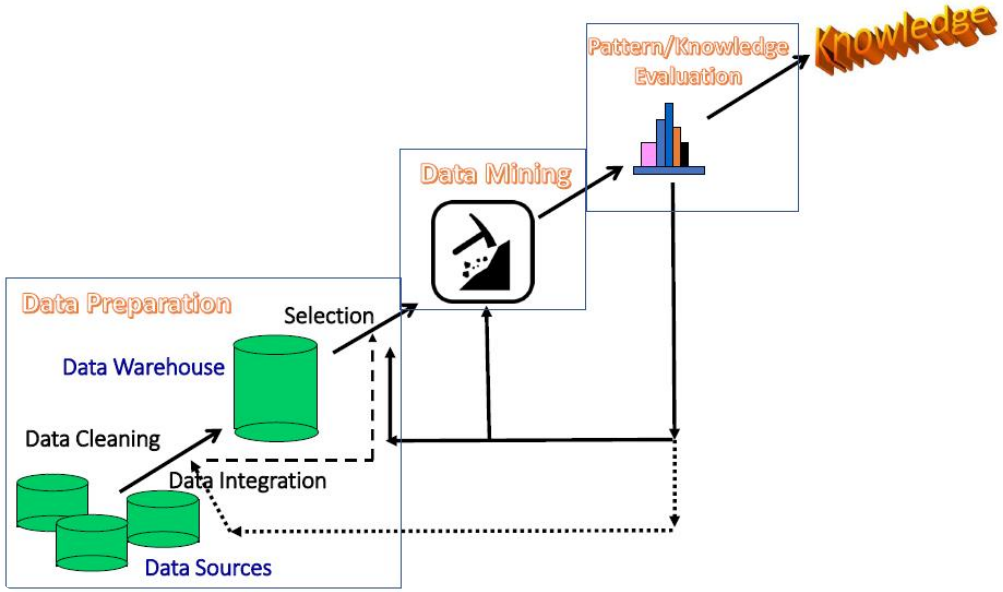
Metodologi Pengembangan: CRISP-DM

Cross-Industry Standard Process for Data Mining

https://medium.com/@sumit.yg/analyzing-seattle-s-airbnb-listings-data-49abdc0977c8



Data Mining: An Essential Step in Knowledge Discovery



- Knowledge Discovery Process
  - Data preparation
    - Data cleaning
    - Data integration
    - Data transformation
    - Data selection
  - Data mining
  - Pattern/model evaluation
  - Knowledge presentation

Diversity of Data Types for Data Mining (I)

- **Structured vs. unstructured data**
  - *Structured*: uniform, record- or table-like structures, defined by data dictionaries, with a fixed set of attributes, each with a fixed set of value ranges and semantic meaning
    - Ex. Data stored in *relational databases*, *data cubes*, *data matrices*, and many *data warehouses*
  - *Semi-structured*: allow a data object to contain a set value, a small set of heterogeneous typed values, or nested structures, or to allow the structure of objects or sub-objects to be defined flexibly and dynamically
  - Data having *certain structures* with clearly defined semantic meaning, such as *transactional data set, sequence data set* (e.g., time-series data, gene or protein data, or Weblog data)
  - *Graph or network data:* A more sophisticated type of semi-structured data set
  - *Unstructured data*: text data and multimedia (*e.g.*, audio, image, video) data
- The real-world data can often be a mixture of structured, semi-structured data and unstructured data

Diversity of Data Types for Data Mining (II)

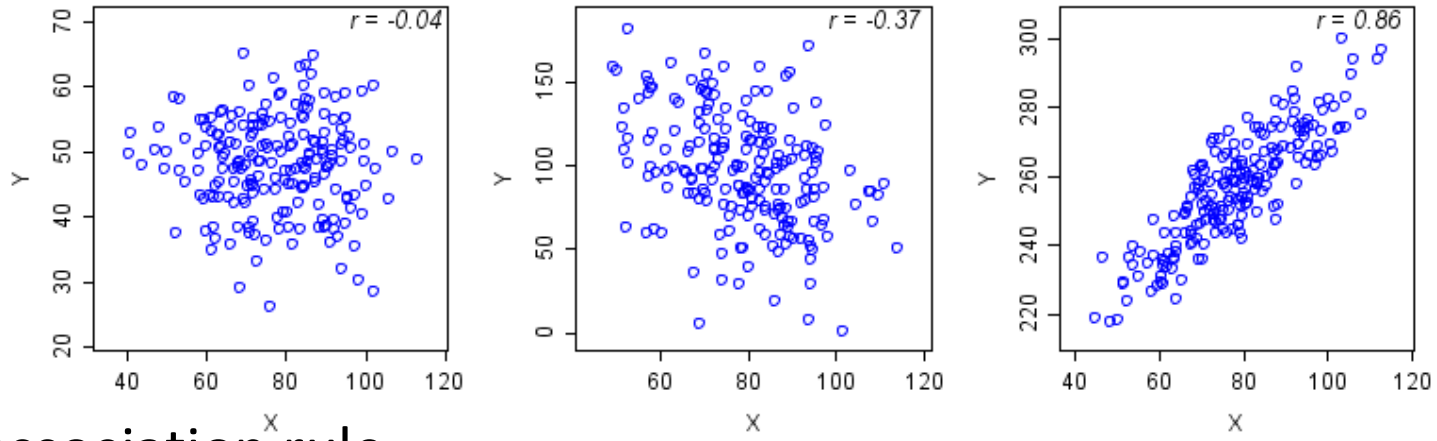
- **Data associated with different applications**
  - Different applications: different data sets and require different data analysis methods
    - Sequence data: *Biological sequences* vs. *shopping transaction sequences*
    - *Time-series:* ordered set of numerical values with equal time interval
    - *Spatial, temporal and spatiotemporal data*
    - Graph and network data: Social networks, computer communication networks, biological networks, and information networks may carry rather different semantics
  - On the same data set, finding different kinds of patterns: require different mining methods
    - Ex. software programs: finding plagiarized modules vs. finding copy-and-paste bugs
- **Stored vs. streaming data**
  - Stored data: Finite, stored in various kinds of large data repositories
  - Streaming data (e.g., video surveillance or remote sensing): Dynamic, constantly coming, infinite, real-time response—posing challenges on effective data mining

Mining Various Kinds of Knowledge

- **Mining Frequent Patterns, Associations, and Correlations**
- **Classification and Regression for Predictive Analysis**
- **Cluster Analysis**
- **Deep Learning**
- **Outlier Analysis**
- **Are All Mining Results Interesting?**

Pattern Discovery: Mining Frequent Patterns, Associations, and Correlations

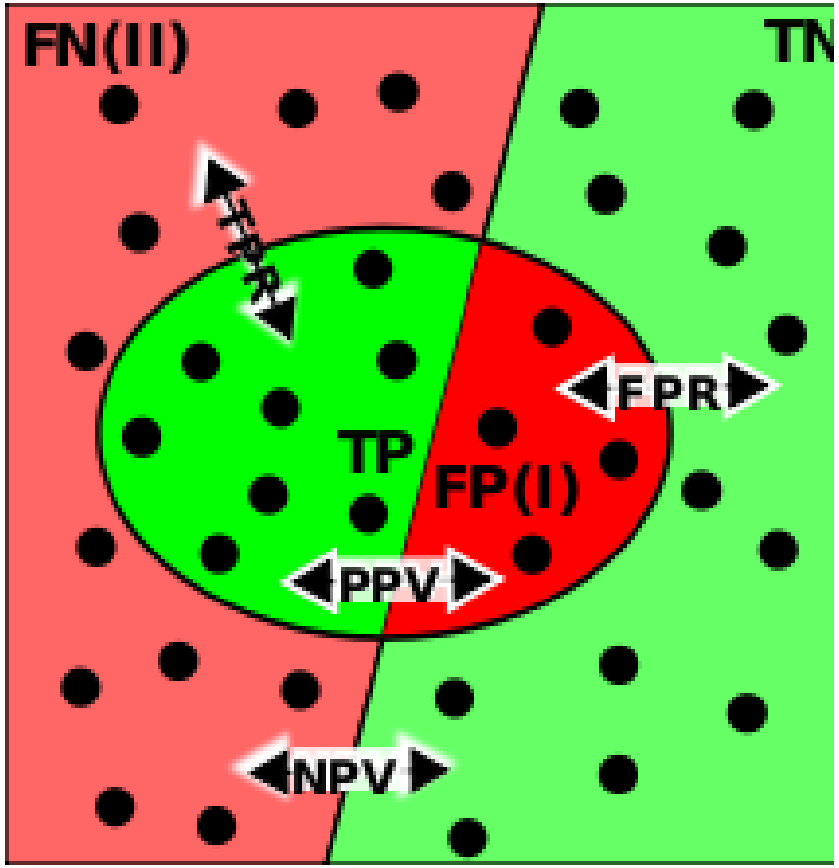
- Frequent patterns (or frequent itemsets)
  - What items are frequently purchased together in your Walmart?
- Association and Correlation Analysis



- A typical association rule
  - Diaper → Beer [0.5%, 75%] (support, confidence)
  - Are strongly associated items also strongly correlated?
- How to mine such patterns and rules efficiently in large datasets?
- How to use such patterns for classification, clustering, and other applications?

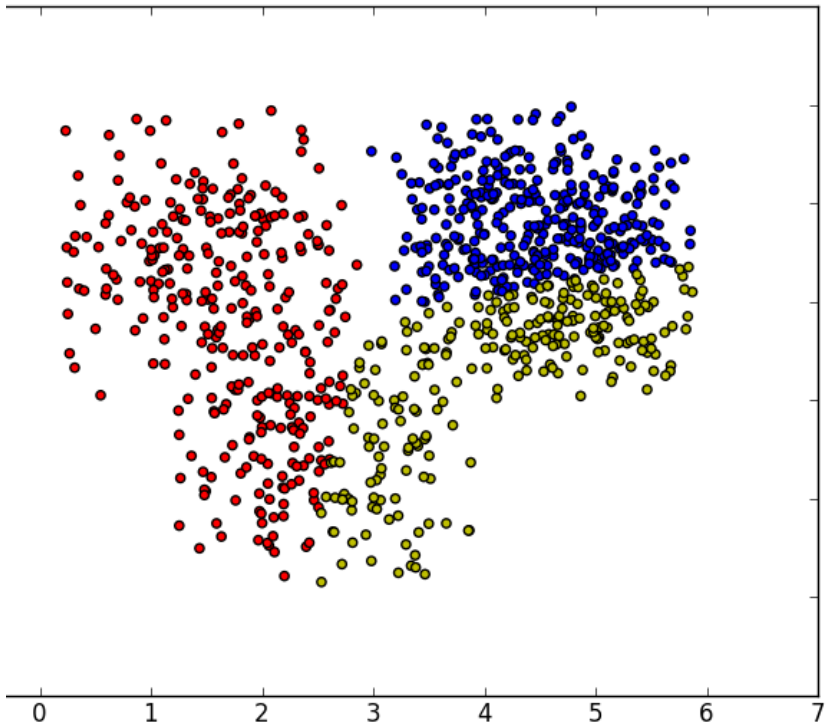
Classification and Regression for Predictive Analysis

- Classification and label prediction
  - Construct models (functions) based on some training examples
  - Describe and distinguish classes or concepts for future prediction
    - Ex. 1. Classify countries based on (climate)
    - Ex. 2. Classify cars based on (gas mileage)
  - Predict some unknown class labels
- Typical methods
  - Decision trees, naïve Bayesian classification, support vector machines, neural networks, rule-based classification, pattern-based classification, logistic regression, …
- Typical applications:
  - Credit card fraud detection, direct marketing, classifying stars, diseases, web-pages, …



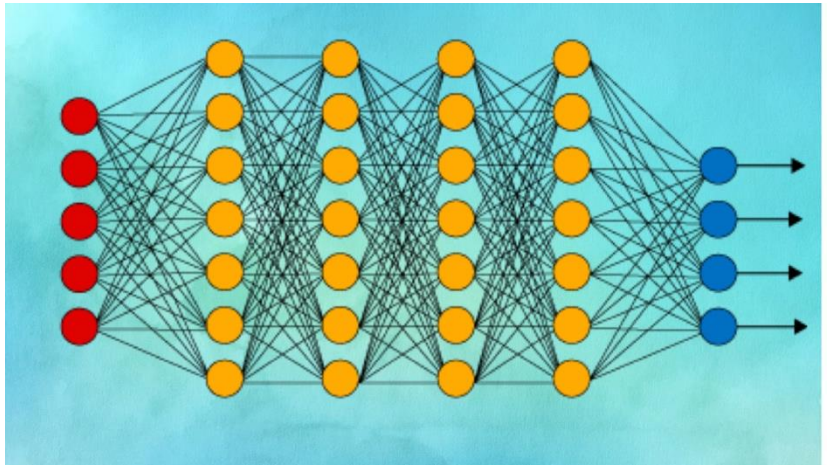
Cluster Analysis

- Unsupervised learning (i.e., Class label is unknown)
- Group data to form new categories (i.e., clusters), e.g., cluster houses to find distribution patterns
- Principle: Maximizing intra-class similarity & minimizing interclass similarity
- Many methods and applications



Deep Learning

- Deep learning: A fast expanding dynamic frontier in machine learning
- Deep learning has developed various *neural network architectures*
  - Feed-forward neural networks
  - Convolutional neural networks
  - Recurrent neural networks
  - Graph neural networks
  - Transformer

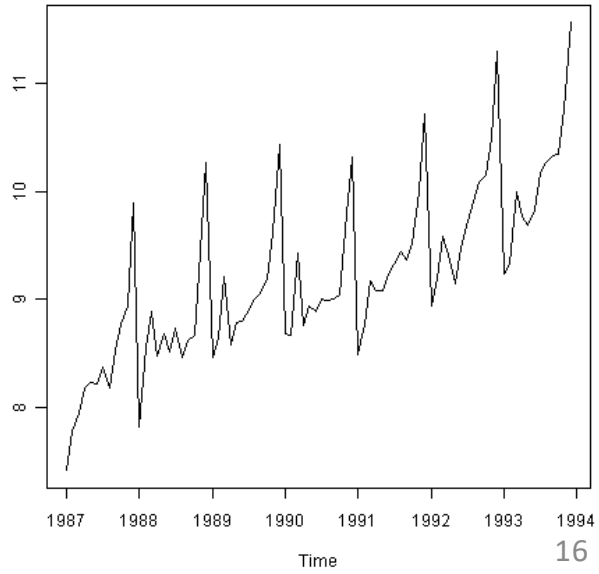


- Deep learning has broad applications in computer vision, natural language processing, machine translation, social network analysis, and so on
- Deep learning has been reshaping a variety of data mining tasks
  - Ex. classification, clustering, outlier detection, and reinforcement learning

Other Data Mining Functions: Time and Ordering: Sequential Pattern, Trend and Evolution Analysis

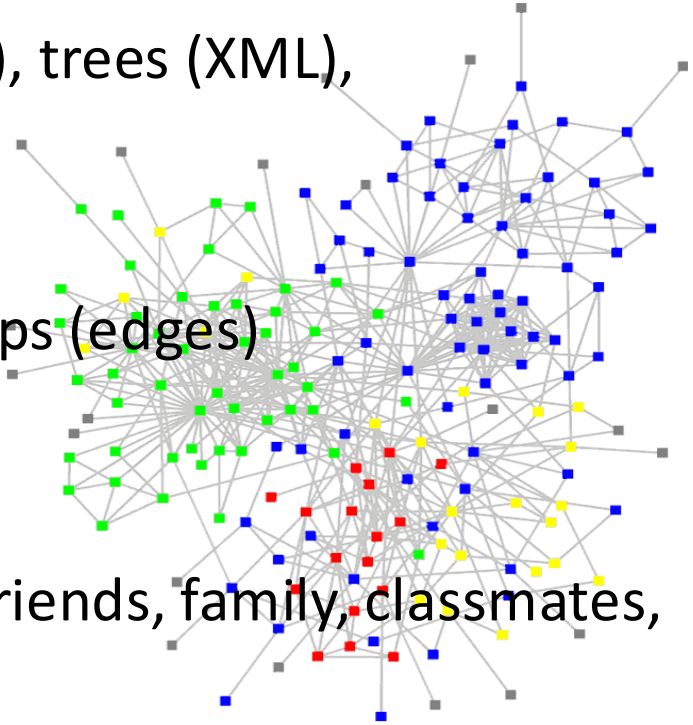
- Sequence, trend and evolution analysis
  - Trend, time-series, and deviation analysis
    - e.g., regression and value prediction
  - Sequential pattern mining
    - e.g., buy digital camera, then buy large memory cards
  - Periodicity analysis
  - Motifs and biological sequence analysis
    - Approximate and consecutive motifs
  - Similarity-based analysis
- Mining data streams
  - Ordered, time-varying, potentially infinite, data streams





Other Data Mining Functions: Structure and Network Analysis

- Graph mining
  - Finding frequent subgraphs (e.g., chemical compounds), trees (XML), substructures (web fragments)
- Information network analysis
  - Social networks: actors (objects, nodes) and relationships (edges)
    - e.g., author networks in CS, terrorist networks
  - Multiple heterogeneous networks
    - A person could be multiple information networks: friends, family, classmates, …
  - Links carry a lot of semantic information: Link mining
- Web mining
  - Web is a big information network: from PageRank to Google
  - Analysis of Web information networks



Data Mining: Confluence of Multiple Disciplines

